

Diffusion Policy · arXiv 2303.04137

Scientific Reviewer

8+3 · seat 5

Strengths + actionable cite hygiene

Sep 21 paper club · VLA Architectures

Spine: chunk-commit-replan (T_p / T_a) — NOT classical planner

Chi et al. · RSS 2023 / IJRR extended

Strengths

Clear problem → method

Multimodality, high-dim sequences,
EBM instability — each gets a design answer

Broad eval

15 tasks · state+vision · sim+real
single + bimanual

Ablations that matter

T_a · latency · pos/vel · encoder
freeze/finetune

Honest IBC analysis

Train instability shown;
last-10 ckpt discipline

Actionable cite hygiene

What to demand when you (or others) cite this paper

1

Shared action interface

Rerun all baselines on same position alphabet — not DP@pos vs LSTM@vel

2

Absolute + hard-task

Don't only quote +46.9%.
Report absolute & hard-task aggregates (Kitchen/Transport)

3

Robomimic bug lore

22 inits not 50 (equalized).
Absolute $\neq$ external 50-init papers

4

Control-module cite

Not a foundation VLM.
Cite for continuous action diffusion/flow + chunks

5

Seed last-10 tables

Release seed-level last-10 vs max-ckpt gap — selection bias still present

6

Verdict

Still the right cite for continuous action diffusion/flow + chunk control

H-pos · position > velocity for Diffusion Policy

Baselines often prefer velocity — shared action interface is the fair cite

POSITION chunks

- Multimodality = which geometric path
 - Less compounding error over T_p
 - Idle / dwell actions natural
- Latency-robust (~ 4 steps, Fig 5)

VELOCITY chunks

- Integrates error open-loop
- CNN oversmooths sharp cmds
 - Idle = zero-vel ambiguity
- Older BC stacks preferred this

Reviewer hygiene: DP@position vs baselines@velocity inflates headline Δ

Empiricist P1: rerun Lift/Square PH state with shared action alphabet

Expect: position $\Delta > 0$ for DP; opposite direction for LSTM-GMM control

Diffuser foil · planning vs control

Janner et al. 2022 vs Chi et al. 2023 — same family, different contract

Diffuser / planning diffusion

- Diffuse joint states + actions
- Long-horizon trajectory plans
- Often open-loop / rare replan
- Explicit dynamics / cost / goals
- Global trajectory consistency

Diffusion Policy (control)

- Diffuse actions ONLY
- Condition on current O_t
- Receding $T_a < T_p$ replan
- Implicit dynamics in denoiser
- Local multimodal $p(A|O)$

§4.5: linear plant + linear demo → optimal denoiser recovers feedback;
multistep $T_p > 1 \Rightarrow$ learner must implicitly learn task dynamics — controller synthesis via BC, not shoot-through planning.

To train the EBM for implicit policy, an InfoNCE-style loss function is used, which equates to the negative log-likelihood of Eq 6:

$$\mathcal{L}_{\text{InfoNCE}} = -\log\left(\frac{e^{-E_\theta(\mathbf{a}, \mathbf{a})}}{e^{-E_\theta(\mathbf{a}, \mathbf{a})} + \sum_{j=1}^{N_{\text{neg}}} e^{-E_\theta(\mathbf{a}, \mathbf{a}^j)}}\right) \quad (7)$$

where a set of negative samples $\{\mathbf{a}^j\}_{j=1}^{N_{\text{neg}}}$ are used to estimate the intractable normalization constant $Z(\mathbf{a}, \theta)$. In practice, the inaccuracy of negative sampling is known to cause training instability for EBMs [Du et al. \(2020\)](#); [Ta et al. \(2022\)](#).

Diffusion Policy and DDPMs sidestep the issue of estimating $Z(\mathbf{a}, \theta)$ altogether by modeling the **score function** [Song and Ermon \(2019\)](#) of the same action distribution in Eq 6:

$$\nabla_{\mathbf{a}} \log p(\mathbf{a}|\mathbf{o}) = -\nabla_{\mathbf{a}} E_\theta(\mathbf{a}, \mathbf{o}) - \underbrace{\nabla_{\mathbf{a}} \log Z(\mathbf{a}, \theta)}_{=0} \approx -E_\theta(\mathbf{a}, \mathbf{o}) \quad (8)$$

where the noise-prediction network $E_\theta(\mathbf{a}, \mathbf{o})$ is approximating the negative of the score function $\nabla_{\mathbf{a}} \log p(\mathbf{a}|\mathbf{o})$ [Liu et al. \(2022\)](#), which is independent of the normalization constant $Z(\mathbf{a}, \theta)$. As a result, neither the inference (Eq 4) nor training (Eq 5) process of Diffusion Policy involves evaluating $Z(\mathbf{a}, \theta)$, thus making Diffusion Policy training more stable.

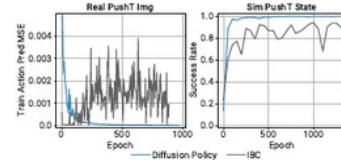


Figure 6. Training Stability. Left: IBC fails to infer training actions with increasing accuracy despite smoothly decreasing training loss for energy function. Right: IBC’s evaluation success rate oscillates, making checkpoint selection difficult (evaluated using policy rollouts in simulation).

4.5 Connections to Control Theory

Diffusion Policy has a simple limiting behavior when the tasks are very simple; this potentially allows us to bring to bear some rigorous understanding from control theory. Consider the case where we have a linear dynamical system, in standard state-space form, that we wish to control:

$$\mathbf{s}_{t+1} = \mathbf{A}\mathbf{s}_t + \mathbf{B}\mathbf{a}_t + \mathbf{w}_t, \quad \mathbf{w}_t \sim \mathcal{N}(\mathbf{0}, \Sigma_w).$$

Now imagine we obtain demonstrations (rollouts) from a linear feedback policy: $\mathbf{a}_t = -\mathbf{K}\mathbf{s}_t$. This policy could be obtained, for instance, by solving a linear optimal control problem like the Linear Quadratic Regulator. Imitating this policy does not need the modeling power of diffusion, but as a sanity check, we can see that Diffusion Policy does the right thing.

In particular, when the prediction horizon is one time step, $T_p = 1$, it can be seen that the optimal denoiser which minimizes

$$\mathcal{L} = \text{MSE}(\epsilon^k, \epsilon_\theta(\mathbf{s}_t, -\mathbf{K}\mathbf{s}_t + \epsilon^k, k)) \quad (9)$$

is given by

$$\epsilon_\theta(\mathbf{s}, \mathbf{a}, k) = \frac{1}{\sigma_k} [\mathbf{a} + \mathbf{K}\mathbf{s}],$$

where σ_k is the variance on denoising iteration k . Furthermore, at inference time, the DDIM sampling will converge to the global minima at $\mathbf{a} = -\mathbf{K}\mathbf{s}$.

Trajectory prediction ($T_p > 1$) follows naturally. In order to predict $\mathbf{a}_{t:t'}$ as a function of $\mathbf{s}_t$, the optimal denoiser will produce $\mathbf{a}_{t:t'} = -\mathbf{K}(\mathbf{A} - \mathbf{B}\mathbf{K})^T \mathbf{s}_t$; all terms involving $\mathbf{w}_t$ are zero in expectation. This shows that in order to perfectly clone a behavior that depends on the state, the learner must implicitly learn a (task-relevant) dynamics model [Subramanian and Mahajan \(2019\)](#); [Zhang et al. \(2020\)](#). Note that if either the plant or the policy is nonlinear, then predicting future actions could become significantly more challenging and once again involve multimodal predictions.

5 Evaluation

We systematically evaluate Diffusion Policy on 15 tasks from 4 benchmarks [Florence et al. \(2021\)](#); [Gupta et al. \(2019\)](#); [Mandlekar et al. \(2021\)](#); [Shafiuallah et al. \(2022\)](#). This evaluation suite includes both simulated and real environments, single and multiple task benchmarks, fully actuated and under-actuated systems, and rigid and fluid objects. We found Diffusion Policy to consistently outperform the prior state-of-the-art on all of the tested benchmarks, with an average success-rate improvement of 46.9%. In the following sections, we provide an overview of each task, our evaluation methodology on that task, and our key takeaways.

5.1 Simulation Environments and datasets

Robomimic [Mandlekar et al. \(2021\)](#) is a large-scale robotic manipulation benchmark designed to study imitation learning and offline RL. The benchmark consists of 5 tasks with a proficient human (PH) teleoperated demonstration dataset for each and mixed proficient/non-proficient human (MH) demonstration datasets for 4 of the tasks (9 variants in total). For each variant, we report results for both state- and image-based observations. Properties for each task are summarized in Tab 3.

Push-T adapted from IBC [Florence et al. \(2021\)](#), requires pushing a T-shaped block (gray) to a fixed target (red) with a circular end-effector (blue). Variation is added by random initial conditions for T block and end-effector. The task requires exploiting complex and contact-rich object dynamics to push the T block precisely, using point contacts. There are two variants: one with RGB image observations and another with 9 2D keypoints obtained from the ground-truth pose of the T block, both with proprioception for end-effector location.

Multimodal Block Pushing adapted from BET [Shafiuallah et al. \(2022\)](#), this task tests the policy’s ability to model multimodal action distributions by pushing two blocks

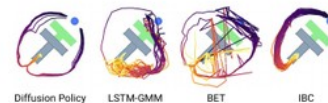


Figure 3. Multimodal behavior. At the given state, the end-effector (blue) can either go left or right to push the block. **Diffusion Policy** learns both modes and commits to only one mode within each rollout. In contrast, both **LSTM-GMM** [Mandekar et al. \(2021\)](#) and **IBC** [Florence et al. \(2021\)](#) are biased toward one mode, while **BET** [Shafiiullah et al. \(2022\)](#) fails to commit to a single mode due to its lack of temporal action consistency. Actions generated by rolling out 40 steps for the best-performing checkpoint.

speculate that there are two primary reasons for this discrepancy: First, action multimodality is more pronounced in position-control mode than it is when using velocity control. Because Diffusion Policy better expresses action multimodality than existing approaches, we speculate that it is inherently less affected by this drawback than existing methods. Furthermore, position control suffers less than velocity control from compounding error effects and is thus more suitable for action-sequence prediction (as discussed in the following section). As a result, Diffusion Policy is both less affected by the primary drawbacks of position control and is better able to exploit position control’s advantages.

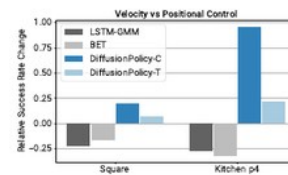


Figure 4. Velocity v.s. Position Control. The performance difference when switching from velocity to position control. While both BCRNN and BET performance decrease, Diffusion Policy is able to leverage the advantage of position and improve its performance.

4.3 Benefits of Action-Sequence Prediction

Sequence prediction is often avoided in most policy learning methods due to the difficulties in effectively sampling from high-dimensional output spaces. For example, IBC would struggle in effectively sampling high-dimensional action space with a non-smooth energy landscape. Similarly, BC-RNN and BET would have difficulty specifying the number of modes that exist in the action distribution (needed for GMM or k-means steps).

In contrast, DDPM scales well with output dimensions without sacrificing the expressiveness of the model, as demonstrated in many image generation applications. Leveraging this capability, Diffusion Policy represents action in the form of a high-dimensional action sequence, which naturally addresses the following issues:

- **Temporal action consistency:** Take Fig 3 as an example. To push the T block into the target from the bottom, the policy can go around the T block from either left or right. However, suppose each action in the sequence is predicted as independent multimodal distributions (as done in BC-RNN and BET). In that case, consecutive actions could be drawn from different modes, resulting in jittery actions that alternate between the two valid trajectories.
- **Robustness to idle actions:** Idle actions occur when a demonstration is paused and results in sequences of identical positional actions or near-zero velocity actions. It is common during teleoperation and is sometimes required for tasks like liquid pouring. However, single-step policies can easily overfit to this pausing behavior. For example, BC-RNN and IBC often get stuck in real-world experiments when the idle actions are not explicitly removed from training.

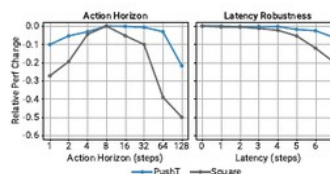


Figure 5. Diffusion Policy Ablation Study. Change (difference) in success rate relative to the maximum for each task is shown on the Y-axis. **Left:** trade-off between temporal consistency and responsiveness when selecting the action horizon. **Right:** Diffusion Policy with position control is robust against latency. Latency is defined as the number of steps between the last frame of observations to the first action that can be executed.

4.4 Training Stability

While IBC, in theory, should possess similar advantages as diffusion policies. However, achieving reliable and high-performance results from IBC in practice is challenging due to IBC’s inherent training instability [Ta et al. \(2022\)](#). Fig 6 shows training error spikes and unstable evaluation performance throughout the training process, making hyperparameter tuning critical and checkpoint selection difficult. As a result, [Florence et al. \(2021\)](#) evaluate every checkpoint and report results for the best-performing checkpoint. In a real-world setting, this workflow necessitates the evaluation of many policies on hardware to select a final policy. Here, we discuss why Diffusion Policy appears significantly more stable to train.

An implicit policy represents the action distribution using an Energy-Based Model (EBM):

$$p_{\theta}(\mathbf{a}|\mathbf{o}) = \frac{e^{-E_{\theta}(\mathbf{a}|\mathbf{o})}}{Z(\mathbf{o}, \theta)} \quad (6)$$

where $Z(\mathbf{o}, \theta)$ is an intractable normalization constant (with respect to $\mathbf{a}$).

Q & A

Right cite · demand hygiene

Diffusion Policy · Sep 21

Diffusion Policy: Visuomotor Policy Learning via Action Diffusion

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Abstract

This paper introduces Diffusion Policy, a new way of generating robot behavior by representing a robot's visuomotor policy as a conditional denoising diffusion process. We benchmark Diffusion Policy across 15 different tasks from 4 different robot manipulation benchmarks and find that it consistently outperforms existing state-of-the-art robot learning methods with an average improvement of 46.9%. Diffusion Policy learns the gradient of the action-distribution score function and iteratively optimizes with respect to this gradient field during inference via a series of stochastic Langevin dynamics steps. We find that the diffusion formulation yields powerful advantages when used for robot policies, including gracefully handling multimodal action distributions, being suitable for high-dimensional action spaces, and exhibiting impressive training stability. To fully unlock the potential of diffusion models for visuomotor policy learning on physical robots, this paper presents a set of key technical contributions including the incorporation of receding horizon control, visual conditioning, and the time-series diffusion transformer. We hope this work will help motivate a new generation of policy learning techniques that are able to leverage the powerful generative modeling capabilities of diffusion models. Code, data, and training details is available diffusion-policy.cs.columbia.edu

Keywords

Imitation learning, visuomotor policy, manipulation

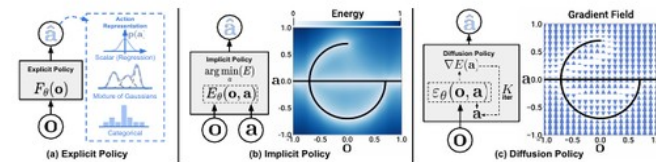


Figure 1. Policy Representations. a) Explicit policy with different types of action representations. b) Implicit policy learns an energy function conditioned on both actions and observation and optimizes for actions that minimize the energy landscape c) Diffusion policy refines noise into actions via a learned gradient field. This formulation provides stable training, allows the learned policy to accurately model multimodal action distributions, and accommodates high-dimensional action sequences.

1 Introduction

Policy learning from demonstration, in its simplest form, can be formulated as the supervised regression task of learning to map observations to actions. In practice however, the unique nature of predicting robot actions — such as the existence of multimodal distributions, sequential correlation, and the requirement of high precision — makes this task distinct and challenging compared to other supervised learning problems.

Prior work attempts to address this challenge by exploring different *action representations* (Fig 1 a) — using mixtures of Gaussians [Mandlekar et al. \(2021\)](#), categorical representations of quantized actions [Shafullah et al. \(2022\)](#), or by switching the *policy representation* (Fig 1 b) — from explicit to implicit to better capture multi-modal distributions [Florence et al. \(2021\)](#); [Wu et al. \(2020\)](#).

In this work, we seek to address this challenge by introducing a new form of robot visuomotor policy that

generates behavior via a “conditional denoising diffusion process” [Ho et al. \(2020\)](#) on robot action space”, **Diffusion Policy**. In this formulation, instead of directly outputting an action, the policy infers the action-score gradient, conditioned on visual observations, for K denoising iterations (Fig. 1 c). This formulation allows robot policies to inherit several key properties from diffusion models — significantly improving performance.

- **Expressing multimodal action distributions.** By learning the gradient of the action score function [Song and Ermon \(2019\)](#) and performing Stochastic Langevin Dynamics sampling on this gradient field, Diffusion policy can express arbitrary normalizable distributions [Neal et al. \(2011\)](#), which includes multimodal action distributions, a well-known challenge for policy learning.